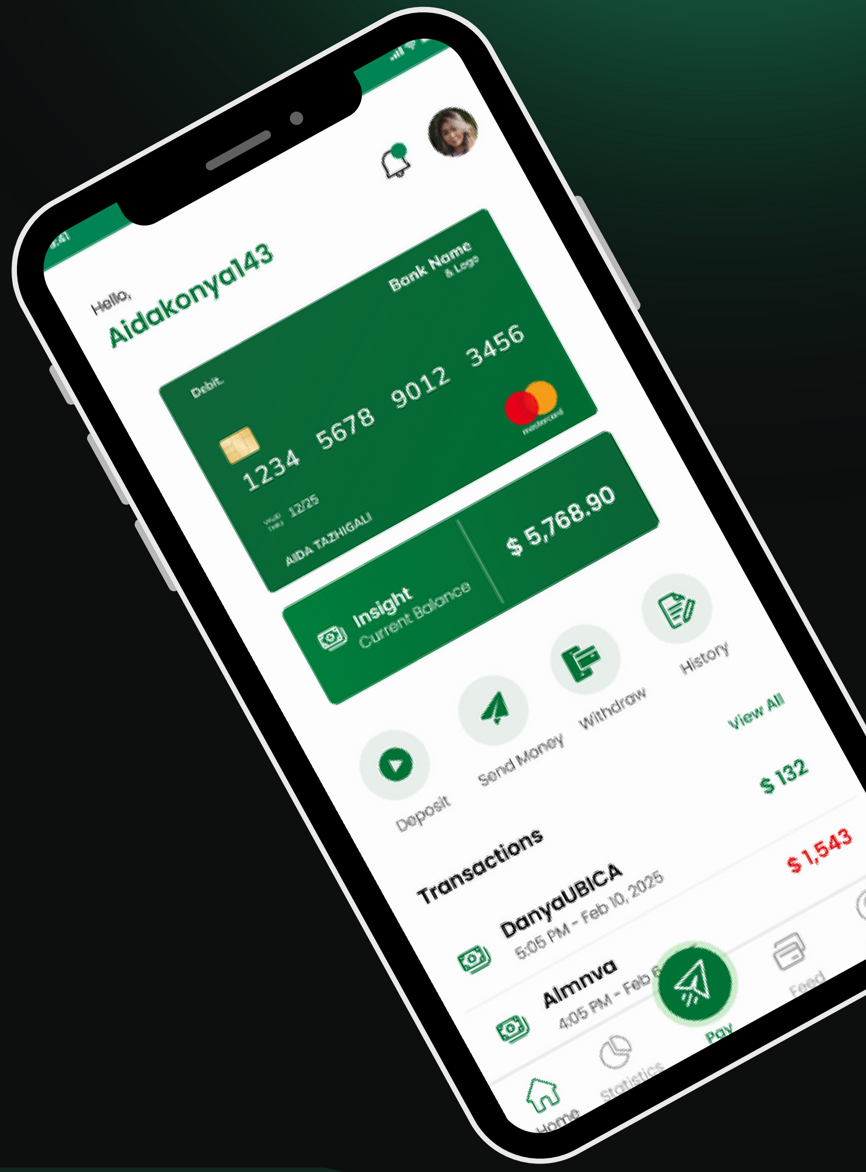
ASTANA IT UNIVERSITY

Diploma Project 2025

Department of Computer Engineering

Development of mobile app to secure transfer of payments through connected bank accounts



SE-2214
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Shynybayev Alikhan

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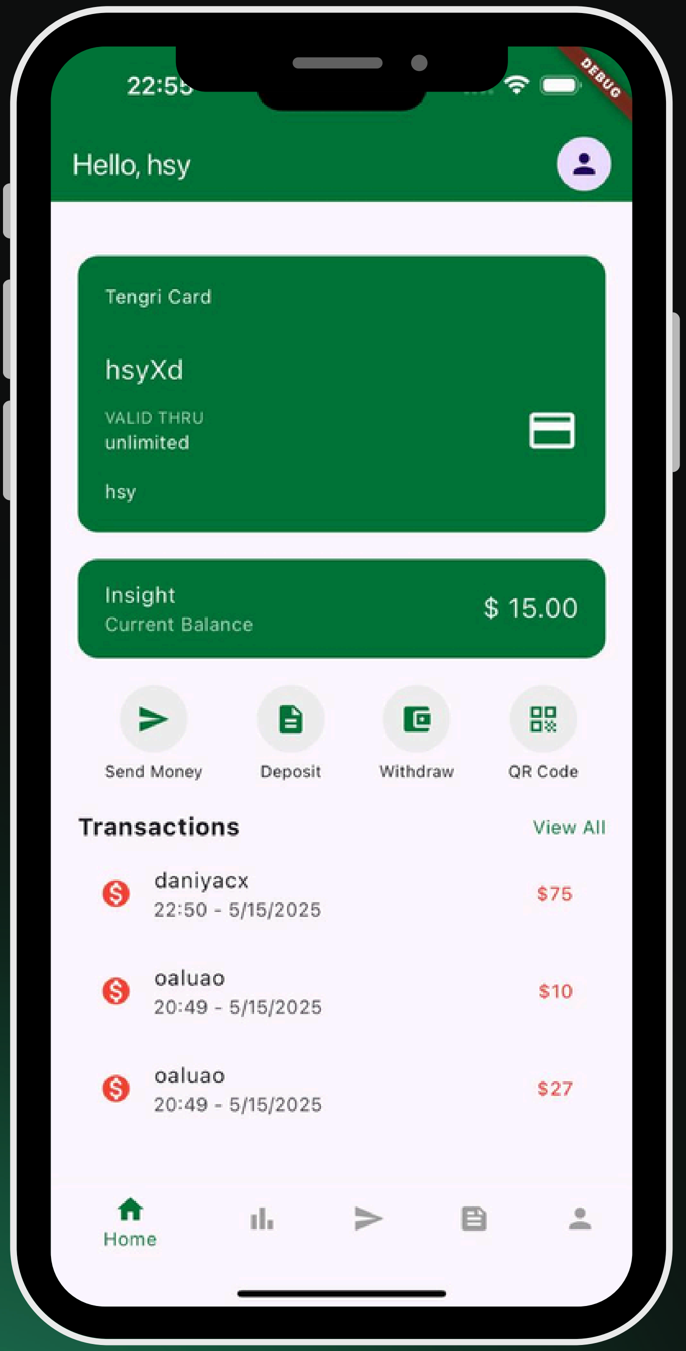
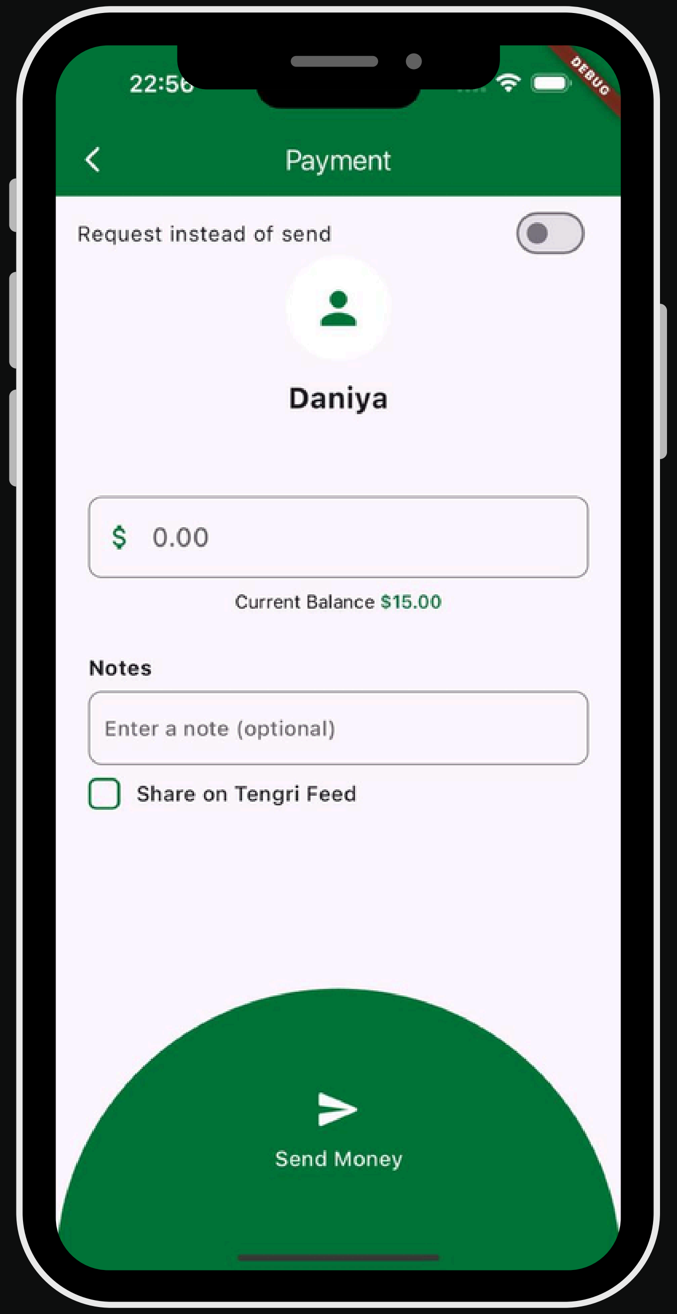
OUTLINE

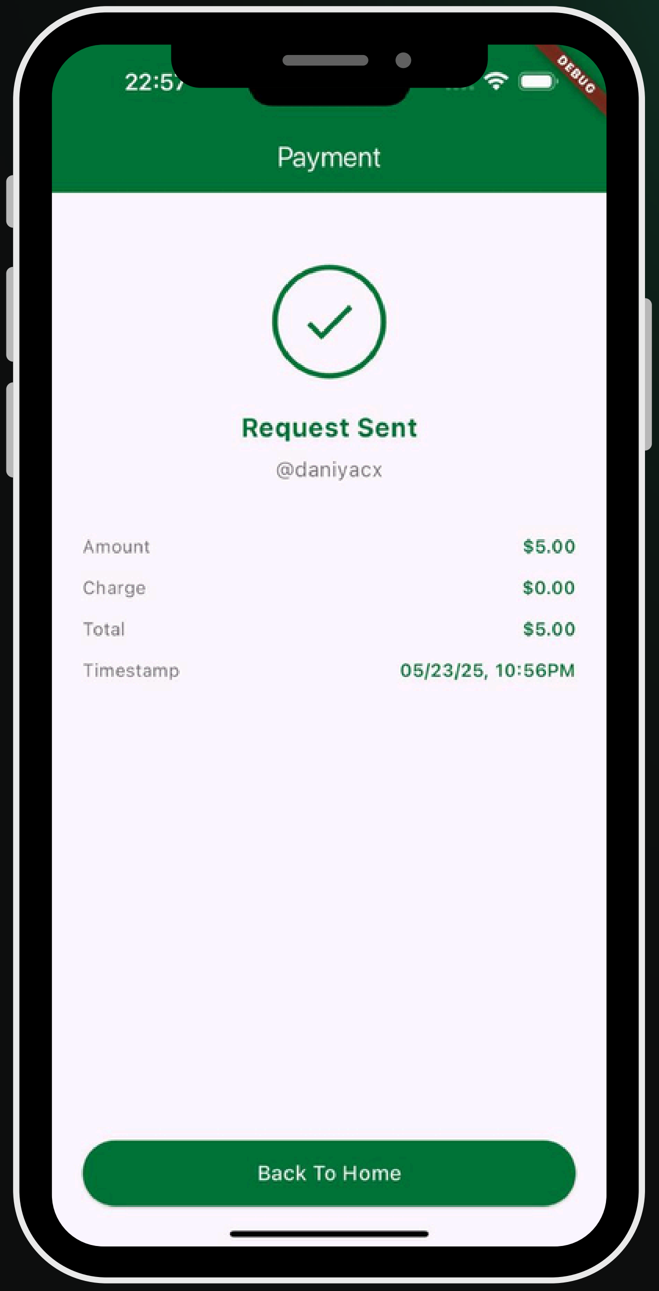
Key Features



Anonymity



No fees



Transaction requests

Introduction

Work Relevance:

Solves digital payment issues—privacy risks, high fees, and complexity.

Novelty:

TengriPay offers anonymous, commission-free Peer to Peer transactions with integrated social features, combining security, privacy, and user engagement in digital payments.

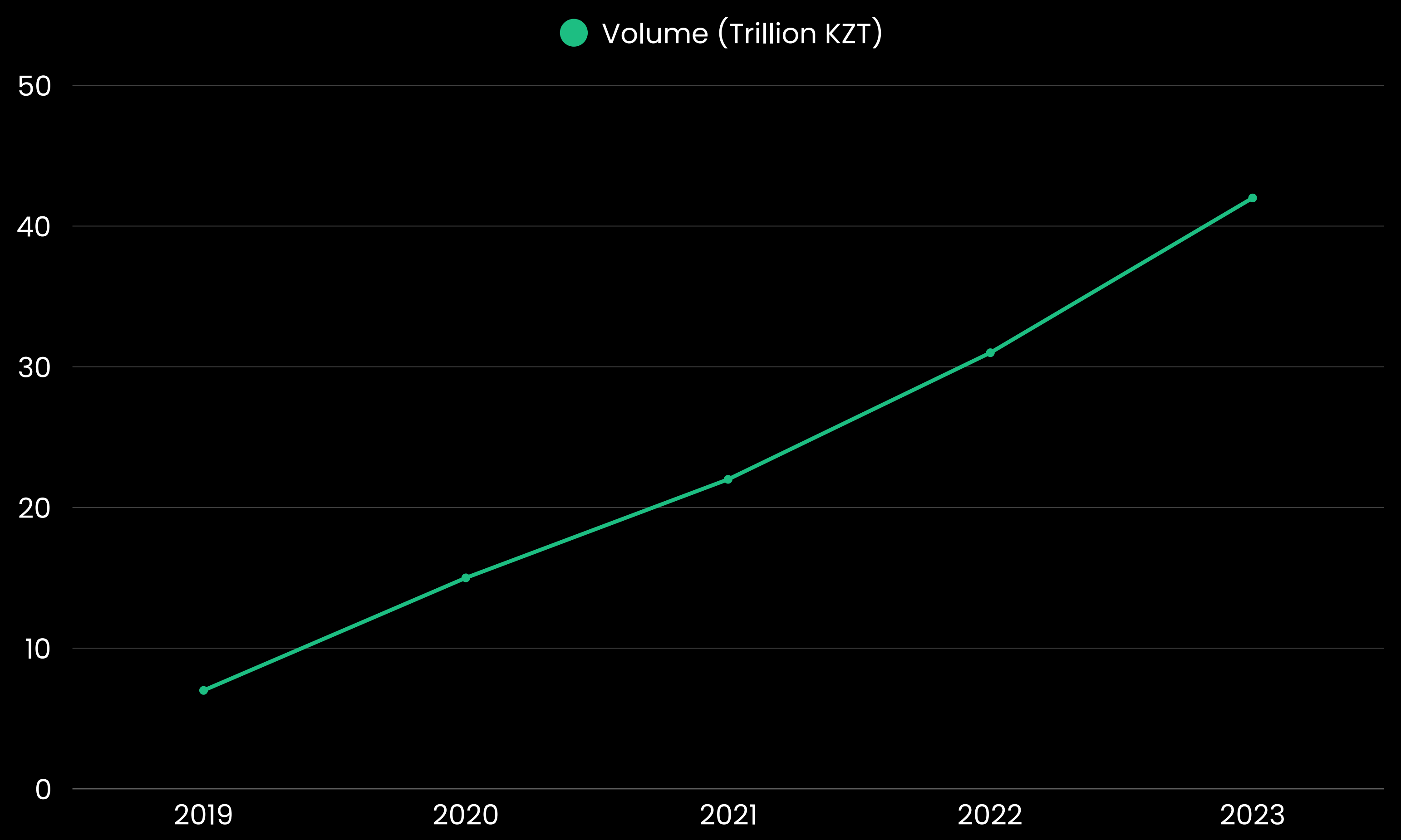
Goal and Objectives:

Develop TengriPay, a secure, user-friendly, cross-platform payment app with social features.

- Secure, encrypted, and anonymous transactions
- Cross-platform development using Flutter
- Integrated social networking for engagement

Literature Review

Propelled by technology, smartphone ubiquity, and state backing, Kazakhstan's digital-payment turnover jumped from 5.9 trn KZT in 2019 to 41.1 trn KZT in 2023, led by Kaspi.kz, Halyk, ForteBank, and Jusan with their mobile banking and e-wallet innovations.



Volume of Cashless Payments in Kazakhstan (2019-2023)

Privacy Concerns in Digital Transactions

Many digital platforms collect user information not only to facilitate transactions but also to create detailed consumer profiles that are used for targeted advertising and analytics [1]

Anonymized Transaction Models

Zero-Knowledge Proofs implemented in privacy-focused cryptocurrencies such as Zcash and Monero allow a party to prove the validity of a transaction without revealing the sender, receiver, or amount [2]

Social Network Integration in Financial Applications

An analysis of 389 million public Venmo transactions found that 10.5 % of notes contained sensitive information, highlighting privacy risks in social payment sharing [3]

Analysis Of Existing Systems

TengriPay: Digital payment app with social features.

Competitors: PayPal, Venmo, Kaspi.kz.

Technology Difference

- TengriPay: Anonymous transfers, social features, no commission.
- PayPal: Widely accepted, but requires personal data and charges fees.
- Venmo: Social payments with fees and offers anonymity.
- Kaspi.kz: All-in-one Kazakhstani platform with commissions.

Features	Paypal	Venmo	Kaspi.kz
Commission-Free transactions for Personal Users	✗	✗	✗
Social Network Functionality	✗	✓	✗
Instant Bank Transfers	✓	✓	✓
Business Accounts with Low Fees	✗	✗	✗
Integrated Digital Wallet	✓	✓	✗

Methodology

Quantitative Research

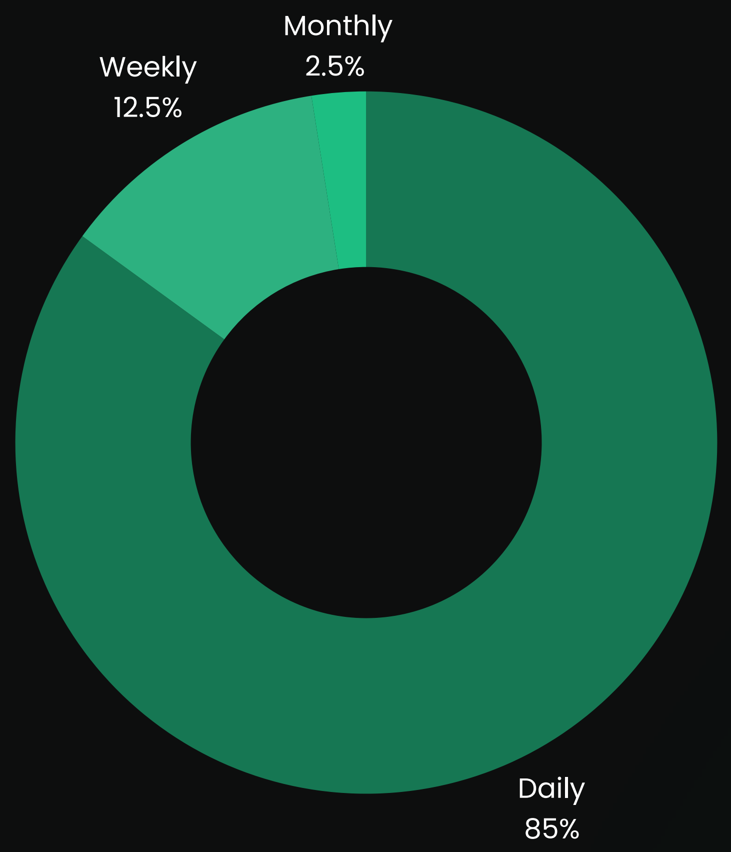
Structured survey is to gather data on user demographics, transaction patterns, privacy concerns, and preferences in digital payment systems. This statistical analysis helped identify trends and inform the design of TengriPay.

Qualitative research

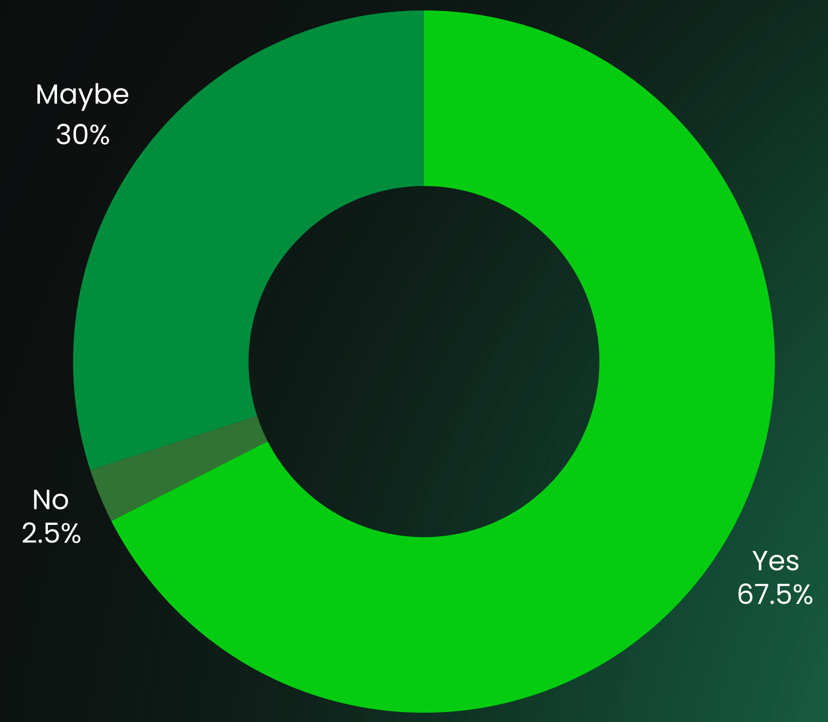
Focus groups and interviews provided deeper insights into user experiences, expectations, and attitudes towards anonymized transactions and social features. This approach ensured a user-focused and innovative payment solution.

Analysis of Collected Quantitative Data

- Most respondents were young adults (18–24), followed by those aged 25–34, showing strong interest in digital payments among youth. Gender was evenly split.
- 77.5% were individual users, with 10% small business owners and 12.5% freelancers, highlighting a primarily personal user base and potential for business features.
- 85% use digital payment apps daily, showing high reliance. Kaspi.kz, Halyk Bank, and Freedom Bank were the top platforms.

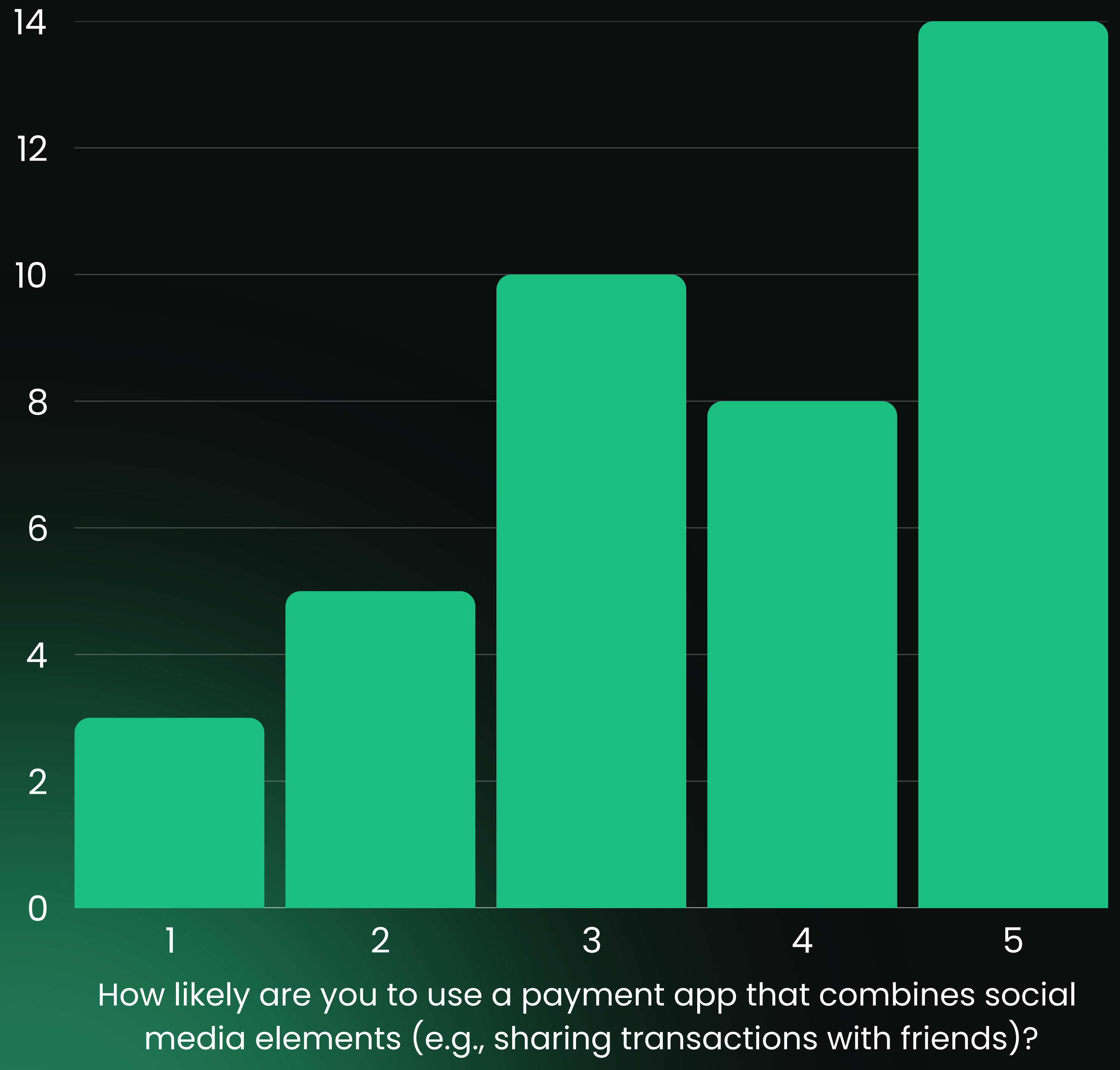


How often do you use digital payment apps?



Would you use a payment app that allows you to send money without revealing personal details to the recipient?

Analysis of Collected Quantitative Data



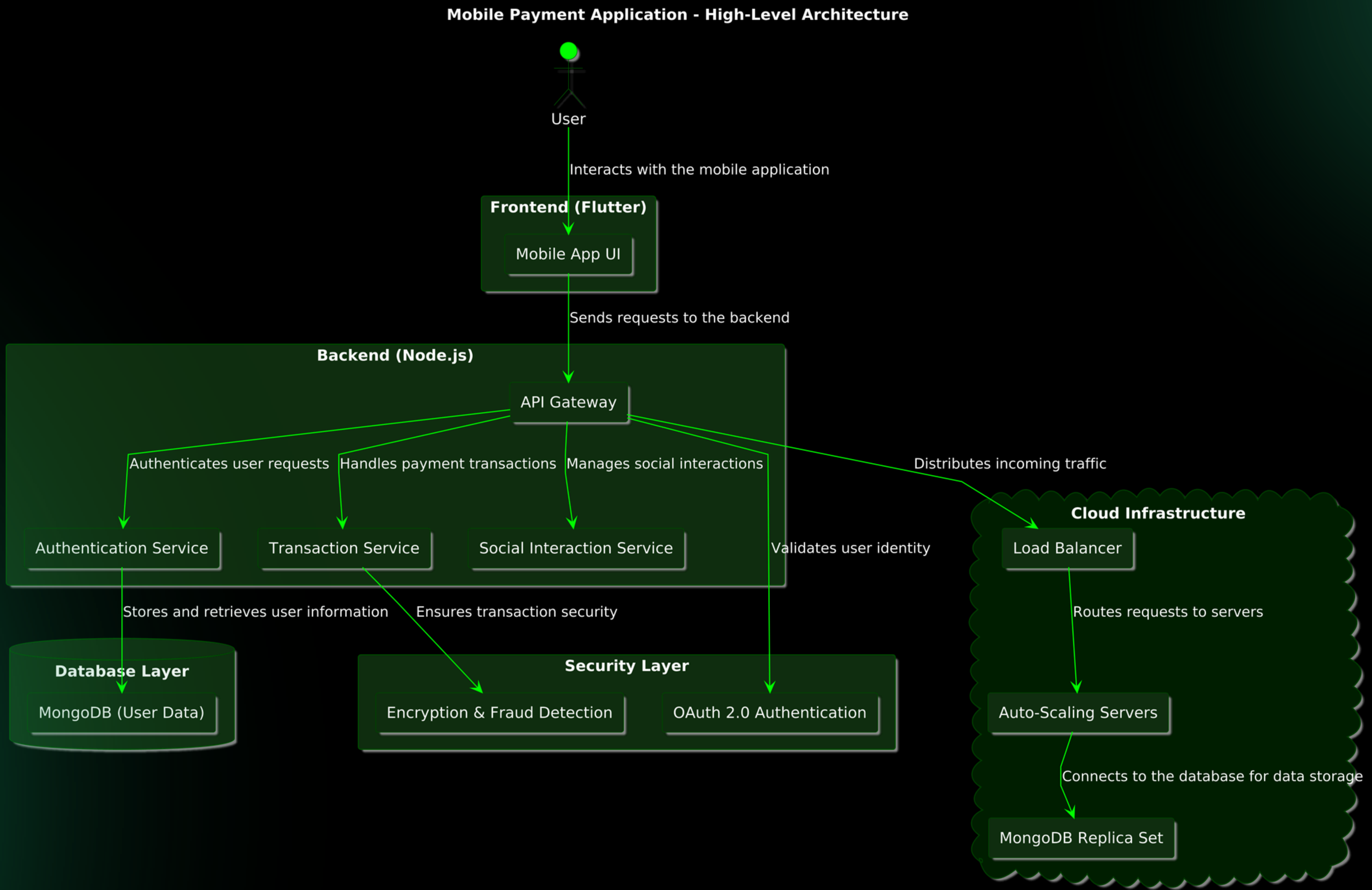
- Key issues included high fees, complex interfaces, slow processing, and security risks like fraud. Many also noted the lack of privacy and anonymity.
- 47.5% considered transaction privacy extremely important, with many preferring apps that support private transfers.
- 35% showed interest in social features like sharing transactions or social media integration, while 25% were neutral.

Analysis of Collected Qualitative Data

Stars (0–3) indicate perceived importance.

Key Functionality	Why Users Want It	General Users	Small-Business Owners
Secure, low-fee payments	Keep costs down, prevent fraud	★★★	★★★
No personal data sharing (usernames)	Anonymity & safety	★★★	★★☆ (needs verification)
Social feed (public / private toggle)	Engagement & optional visibility	★★☆	★★☆ (for promos)
Request / split payments	Convenience & clarity	★★★	★★☆ (tips, invoices)
Business tools (sales, accounting)	Streamline ops & marketing	—	★★★

Software Architecture



Customer Journey Map

1 AWARENESS

- Sees an ad or hears about TengriPay via social media or friends.
- Hesitation due to privacy or unknown brand
- Emphasizes anonymity and privacy in ad messaging

2 CONSIDERATION

- Reviews features and testimonials
- Concerns about hidden fees or identity exposure
- Promotes commission-free transfers and username-based accounts

3 PURCHASE

- Registers an account with a unique username (no personal data sharing).
- Worries about sharing real info or linking a bank card
- Allows account creation with just a username and secure linking

4 RETENTION

- Uses the app for secure P2P transfers with friends and small businesses.
- Shares payment notes in the social feed (with privacy toggles).
- Contacts in-app support if issues arise.

5 ADVOCACY

- Advocates for the app by sharing it with friends or posting about it in the social feed.
- Only promotes if it feels trustworthy and useful
- Gains trust by addressing core pains and offering unique privacy and social benefits

Technical Solutions

Backend

TengriPay's backend uses Node.js microservices (auth, user, feed, transaction) and MongoDB for secure, scalable cloud data storage.

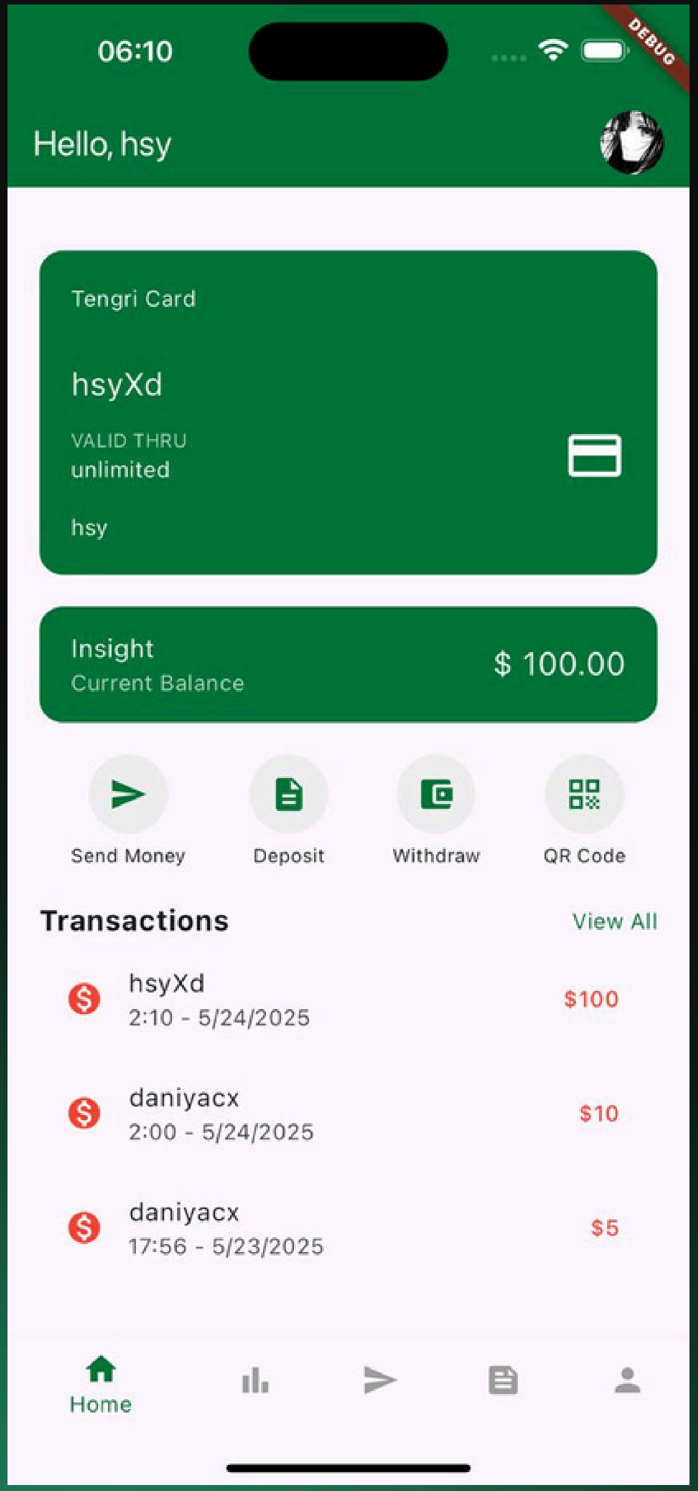
- Auth handles JWT-based login with BCrypt.
- User manages profiles and friend lists.
- Feed provides a social timeline via REST APIs.
- Transaction enables peer-to-peer (P2P) transfers, top-ups, and withdrawals with strict validation and atomic MongoDB operations.

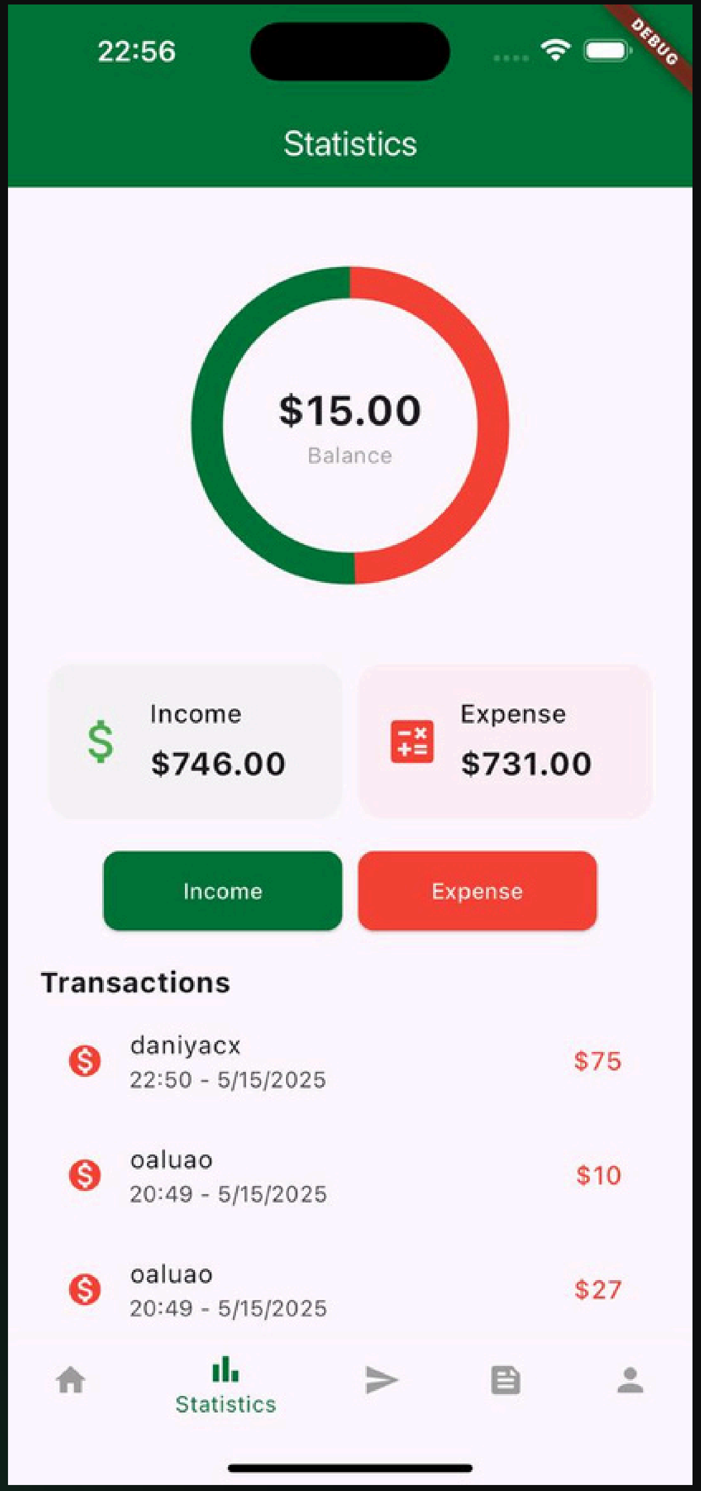
Frontend

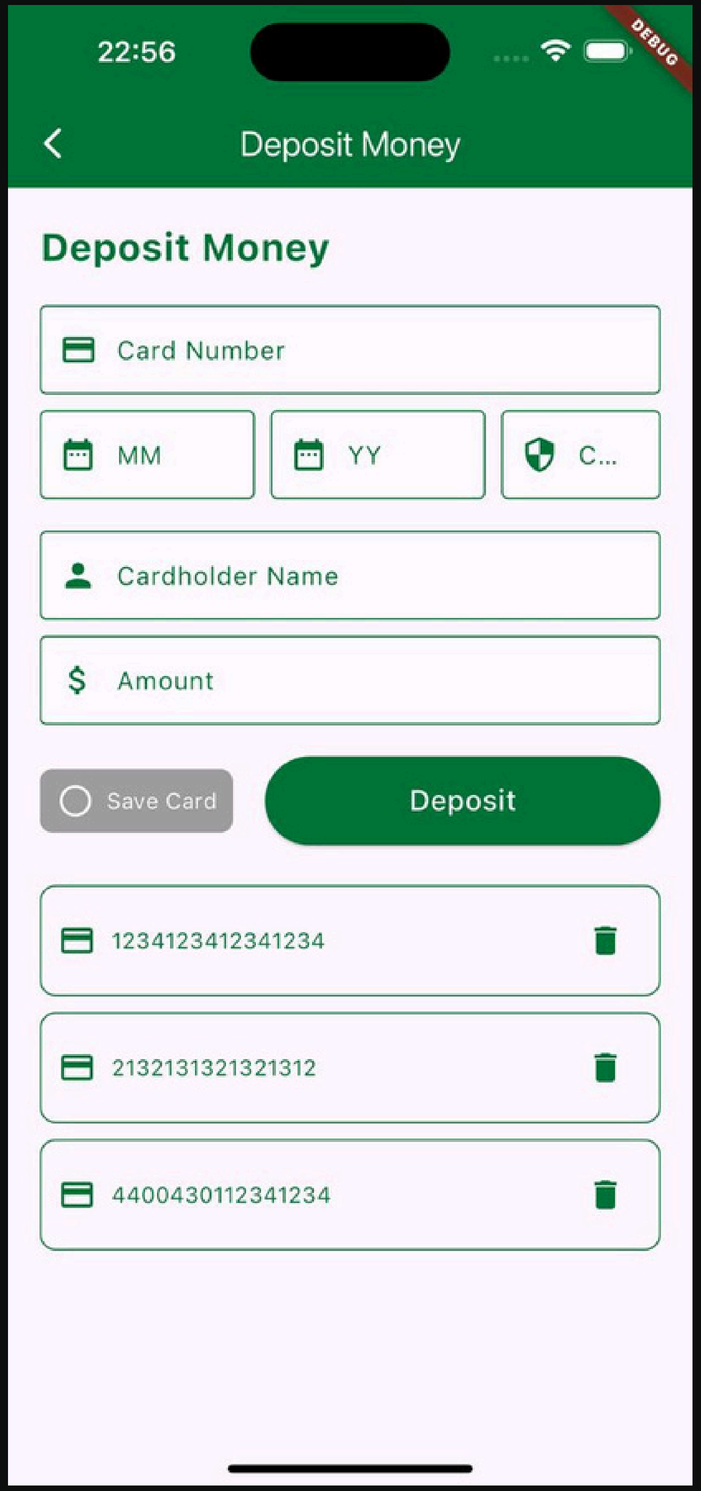
TengriPay's frontend is built with Flutter, ensuring a smooth and responsive user experience across devices.

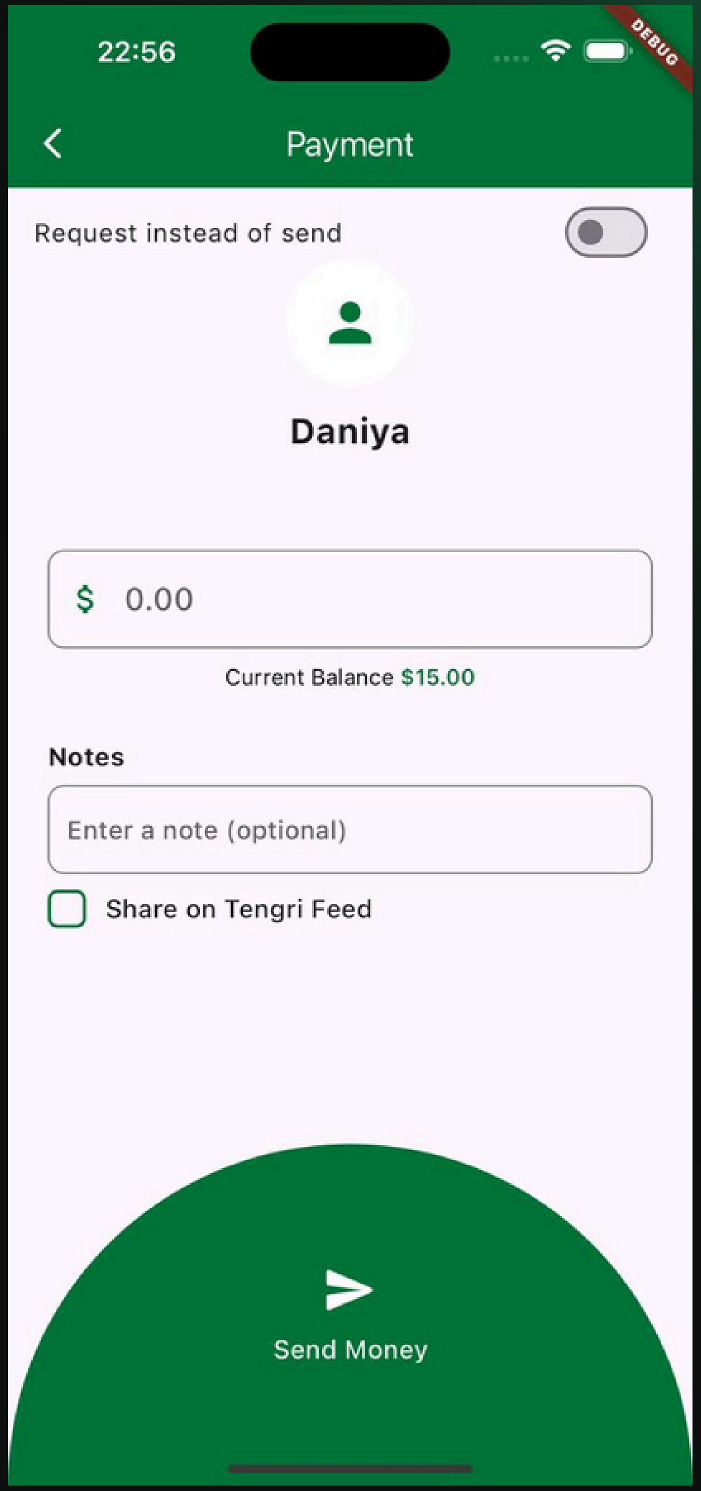
- Started with Figma wireframes for layout and UX design.
- Key modules: auth, wallet, social feed, transactions, and statistics.
- Maintained visual consistency using a green theme, minimalist icons, and adaptive card layouts.
- Final UI was optimized through real-device testing, refining padding, element sizes, and touch areas for better usability.

Results


06:10
DEBUG
Hello, hsy
Tengri Card
hsyXd
VALID THRU
unlimited
hsy
Insight
Current Balance
$ 100.00
Send Money
Deposit
Withdraw
QR Code
Transactions
View All
hsyXd
2:10 - 5/24/2025
$100
daniyacx
2:00 - 5/24/2025
$10
daniyacx
17:56 - 5/23/2025
$5
Home


22:56
DEBUG
Statistics
$15.00
Balance
Income
$746.00
Expense
$731.00
Income
Expense
Transactions
daniyacx
22:50 - 5/15/2025
$75
oaluao
20:49 - 5/15/2025
$10
oaluao
20:49 - 5/15/2025
$27
Statistics


22:56
DEBUG
Deposit Money
Deposit Money
Card Number
MM
YY
C...
Cardholder Name
Amount
Save Card
Deposit
1234123412341234
2132131321321312
4400430112341234


22:56
DEBUG
Payment
Request instead of send
Daniya
$ 0.00
Current Balance $15.00
Notes
Enter a note (optional)
Share on Tengri Feed
Send Money

Reflection

Alua Alamanova

Frontend Developer & Database Designer

Alikhan Shynybayev

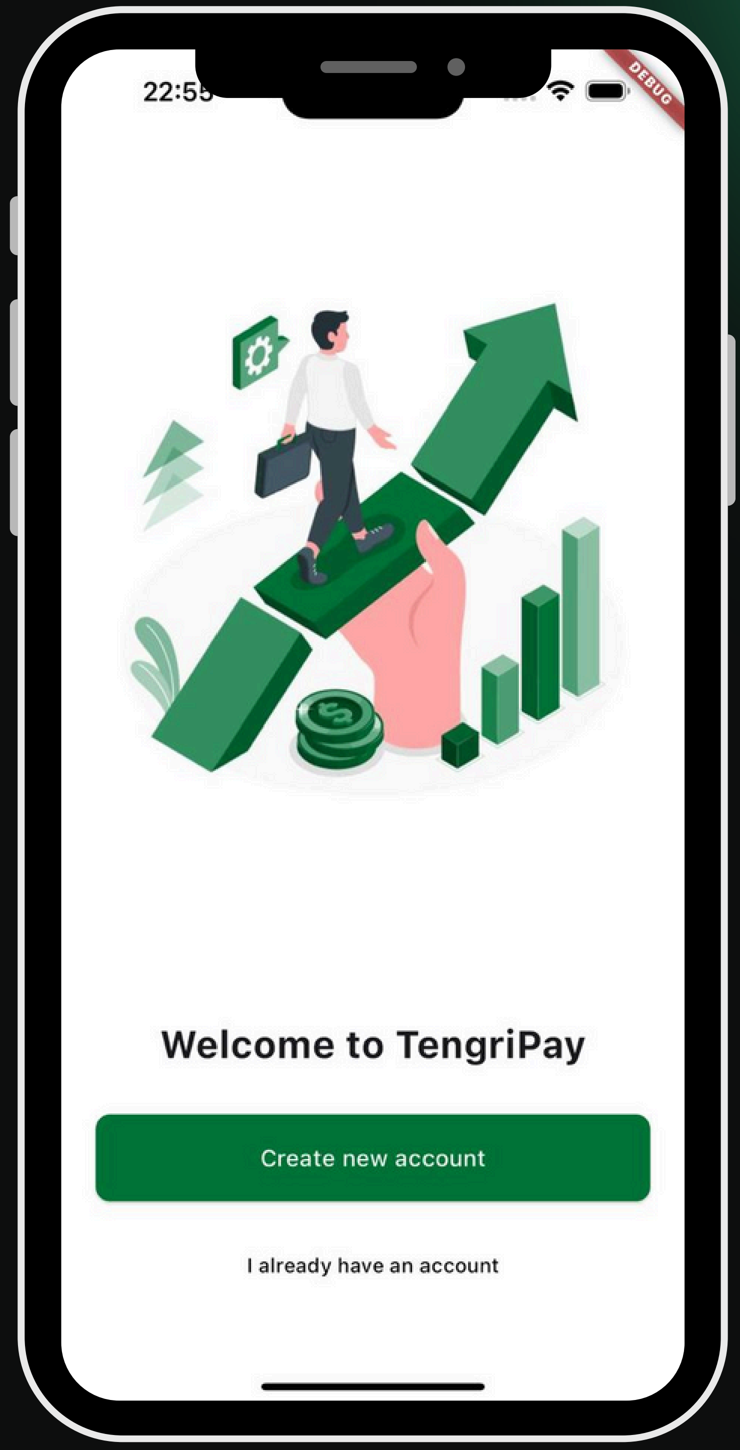
Backend Developer & System Architect

Daniya Aubakirova

Frontend Developer & Database Designer

Conclusion

This study culminated in the creation of TengriPay - a mobile application that unites commission-free peer-to-peer transfers, username based anonymity, and optional social sharing in a secure, cross-platform environment. Informed by user surveys and literature highlighting challenges in privacy, fees, and user engagement, TengriPay offers a flexible, user-centric solution that addresses these gaps. By combining robust security architecture with real-time features and user-controlled privacy, this study demonstrates that secure, private, and socially flexible payments are both technically viable and responsive to the needs of Kazakhstan's digital payment community.



References

[1] V. Fgbou, “Digital analytics and financial security control of socially significant organizations,” —, 2022, DOI: 10.12737/1863937.

[2] W. Yong, C. Lijie, W. Yifan, and W. Qiancheng, “Efficient and secure confidential transaction scheme based on commitment and aggregated zero-knowledge proofs,” Journal of Cyber Security Technology, pp. 1–21, 2024, DOI: 10.1080/23742917.2024.2336634.

[3] R. Tandon, P. Charnsethikul, I. Arora, D. Murthy, and J. Mirkovic, “I know what you did on venmo: Discovering privacy leaks in mobile social payments,” Proceedings on Privacy Enhancing Technologies, no. 3, pp. 200–221, 2022, DOI: 10.56553/POPETS-2022-0069.

[4] M. Hasan, O. Talukder, R. K. Saju, and M. Lysuzzaman. (2025) KYC-driven cryptocurrency wallet: A solution to regulatory compliance. Accessed 22 May 2025. [Online]. Available: https://www.researchgate.net/publication/388449132 KYC-DRIVEN CRYPTOCURRENCY WALLET A SOLUTION TO REGULATORY COMPLIANCE

Thank You